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N + - Network Fundamentals & Building Networks.

Layer-3-Switch-Inter-VLAN-Routing.

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TASK 1.

❖ On a Cisco Layer 3 switch, create three VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support', VLAN 30 for 'Marketing') and assign at least one interface to each VLAN.

- **Create Three VLANs on a Layer 3 Switch:**

VLAN Configuration:

```
enable
configure terminal

vlan 10
name Sales
exit

vlan 20
name Support
exit

vlan 30
name Marketing
exit
```

Assign Interfaces to VLANs:

```
interface fa0/1
switchport mode access
switchport access vlan 10
exit

interface fa0/2
switchport mode access
switchport access vlan 20
exit
```

interface fa0/3
switchport mode access
switchport access vlan 30
exit

Verify Configuration:

show vlan brief

Result: Three VLANs Sales, Support, and Marketing were successfully created, with one interface assigned to each VLAN.

TASK 2.

❖ Configure Switch Virtual Interfaces (SVIs) for each VLAN you created by assigning them unique IP addresses (e.g., 192.168.10.1/24 for VLAN 10, etc.) and ensure each SVI is in 'up/up' state, Use the 'interface vlan' command to create SVIs and 'no shutdown' to enable them.

- **Configure Switch Virtual Interfaces (SVIs):**

SVI Configuration:

enable
configure terminal

ip routing

interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit

interface vlan 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit

Verify SVI Status:

show ip interface brief

Result: SVIs for VLANs 10, 20, and 30 are configured with unique IP addresses and should show up/up when each VLAN has an active port.

TASK 3.

❖ Enable IP routing on the Layer 3 switch so that devices in different VLANs can communicate with each other without using an external router, Use the 'ip routing' global configuration command.

- **Enable IP Routing on Layer 3 Switch:**

Configuration:

enable
configure terminal
ip routing

Verification:

show ip route

Result: IP routing is enabled, allowing devices in different VLANs to communicate directly through the Layer 3 switch without an external router.

TASK 4.

❖ Verify inter-VLAN routing by connecting two PCs (one in VLAN 10 and one in VLAN 20) and pinging between them. Capture and submit the ping results to show successful communication.

➢ Verify Inter-VLAN Routing:

PC Configuration:

- PC1 (VLAN 10): IP 192.168.10.2, Gateway 192.168.10.1
- PC2 (VLAN 20): IP 192.168.20.2, Gateway 192.168.20.1

Ping Test:
From PC1, open Command Prompt and enter:

ping 192.168.20.2

Result: The ping is successful, confirming that PCs in VLAN 10 and VLAN 20 can communicate through the Layer 3 switch.

Screenshot:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	PC3	ICMP		0.000	N	0
	Successful	PC3	PC6	ICMP		0.000	N	1
	Successful	PC2	PC7	ICMP		0.000	N	2

Figure : Successful Inter-VLAN Communication Between PCs Using ICMP in Cisco Packet Tracer.

TASK 5.

❖ Refactor your VLAN and SVI configuration so that VLAN 30 uses a different subnet mask (e.g., /25 instead of /24). Explain what changes you made and how this affects communication with the other VLANs.

- **Change VLAN 30 Subnet Mask to /25:**

Updated SVI Configuration:

enable
configure terminal

interface vlan 30
ip address 192.168.30.1 255.255.255.128
no shutdown
exit

Changes Made:

- **VLAN 30 subnet mask changed from /24 to /25.**
- **New subnet: 192.168.30.0/25.**
- **Valid host range: 192.168.30.1 – 192.168.30.126.**

Effect on Communication:

VLAN 30 can still communicate with VLAN 10 and VLAN 20 through Layer 3 routing. However, devices in VLAN 30 must use the new /25 subnet mask and remain within its valid IP range.

Result: VLAN 30 now uses a /25 subnet mask, while inter-VLAN communication remains possible.